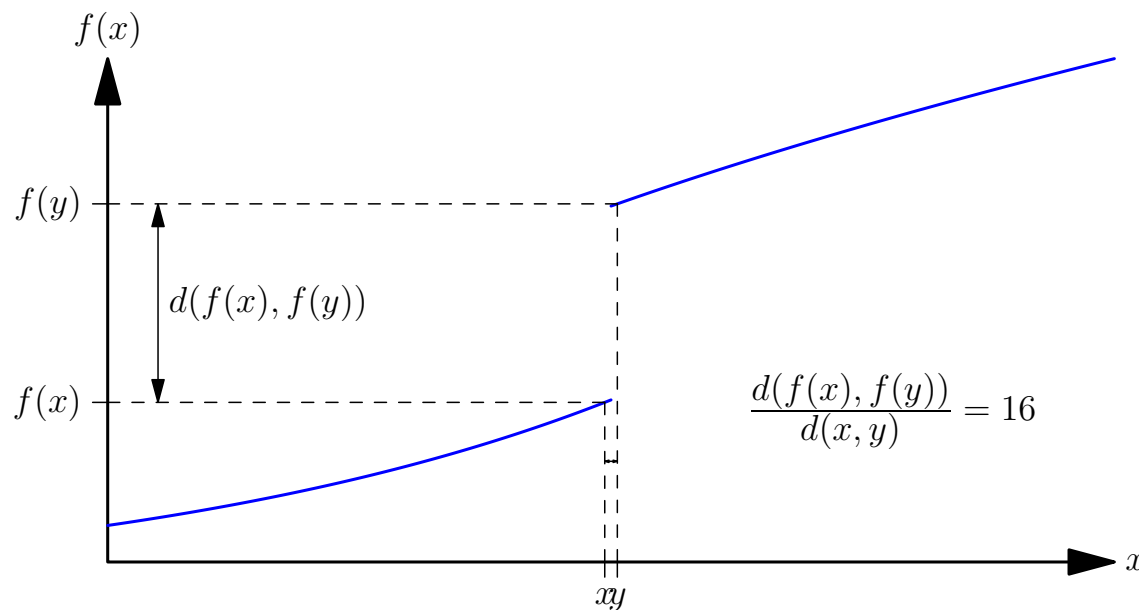


Mathematics of Machine Learning

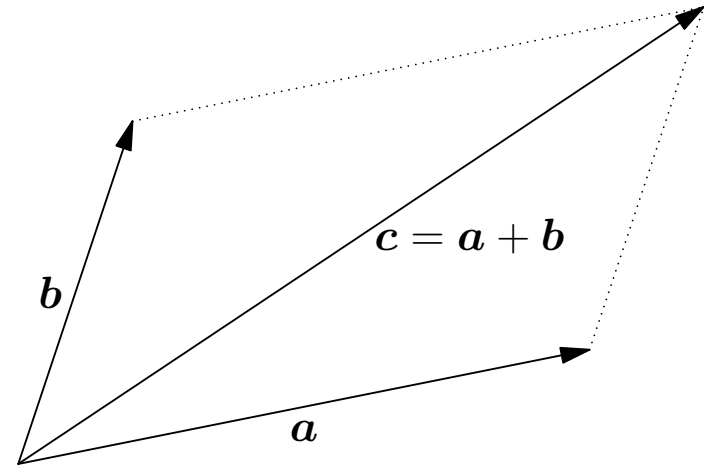
Vector Spaces



Vectors, metric spaces, norms

Outline

1. **Vector Spaces**
2. Metrics (distances)
3. Norms



Matrices, Vectors and All That

- The language of machine learning is mathematics■
- Sometimes we draw pretty pictures to explain the mathematics■
- Much of the mathematics we will use involves vectors, matrices and functions■
- You need to master the language of mathematics, otherwise you won't understand the algorithms■
- I'm going to spend this lecture and the next revising the mathematics you need to know■(but I'm going to use a slightly posher language than you are probably used to)■

Scalars (Fields)

- Vector spaces involve **fields** (numbers)—aka **scalars**
- These are quantities we can add together ($a + b$) and multiply together ($a \times b$)
- Formally they form an Abelian group under addition with an identity 0 and excluding 0 an Abelian group under multiplication and they are distributive

$$a \times (b + c) = a \times b + a \times c$$

- Although this sounds rather daunting don't panic. They behave like numbers. The field might be rational numbers, reals, complex numbers or something a bit more exotic (not the integers: 2 has no integer inverse)—but we will almost always consider reals

Vectors

- We often work with objects with many components (features)■
- To help handle this we will use vector notation■

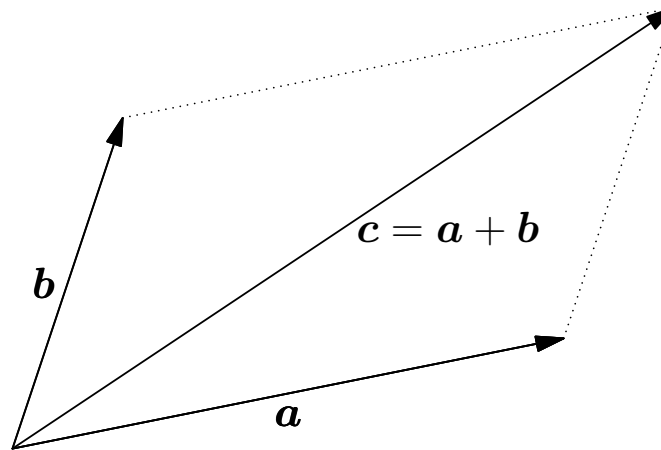
$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \blacksquare$$

- We represent vectors by bold symbols■
 - All our vectors are column vectors by default■
 - We treat them as $n \times 1$ matrices■
- We write row vectors as transposes of column vectors

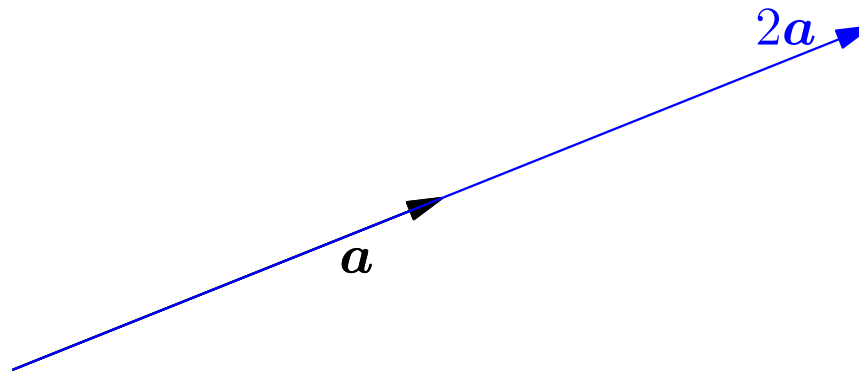
$$\mathbf{y}^T = (y_1, y_2, \dots, y_n) \blacksquare$$

Basic Vector Operations

- The basic vector operations are adding



- multiplying by a scalar (a number)



Vector Space

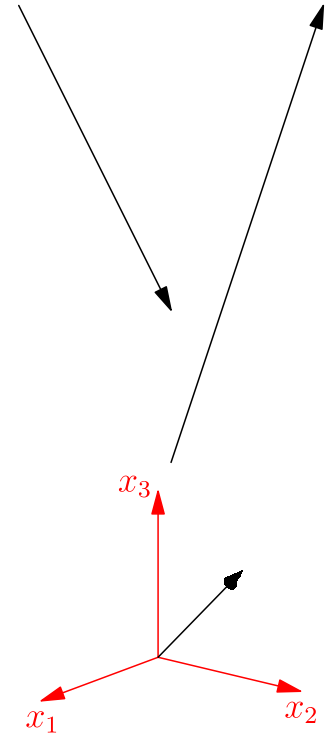
- A vector space, $\mathcal{V}$, is a set of vectors which satisfies
 1. if $\mathbf{v}, \mathbf{w} \in \mathcal{V}$ then $a\mathbf{v} \in \mathcal{V}$ and $\mathbf{v} + \mathbf{w} \in \mathcal{V}$ (closure)■
 2. $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$ (commutativity of addition)■
 3. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ (associativity of addition)■
 4. $\mathbf{v} + \mathbf{0} = \mathbf{v}$ (existence of additive identity $\mathbf{0}$)■
 5. $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$ (existence of additive inverse $-\mathbf{v}$)■
 6. $1\mathbf{v} = \mathbf{v}$ (existence of multiplicative identity 1)■
 7. $a(b\mathbf{v}) = (ab)\mathbf{v}$ (associativity)■
 8. $a(\mathbf{v} + \mathbf{w}) = a\mathbf{v} + a\mathbf{w}$ (distributive properties)■
 9. $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$ ■

(You don't need to remember these)■

- Just from these properties we can deduce other properties■

$$\mathbb{R}^n$$

- When we first learn about vectors we think of them as arrows in 3-D space■
- If we centre them all at the origin then there is a one-to-one correspondence between vectors and points in space■
- We call this vector space $\mathbb{R}^3$ ■
- Any set of quantities $\mathbf{x} = (x_1, x_2, \dots, x_n)^\top$ which satisfy the axioms above form a vector space $\mathbb{R}^n$ ■
- Of course, we can't so easily draw pictures of high-dimensional vectors■

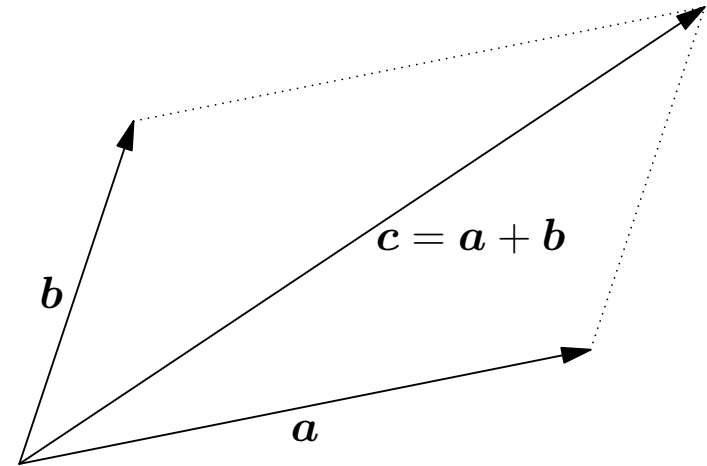


Other Vector Spaces

- Any objects that satisfy the axioms of a vector space are *vectors* — not just $v \in \mathbb{R}^n$
- Matrices satisfy all the conditions of a vector space
- Infinite sequences form a vector space
- Functions form a vector space
 - ★ Let $C(a,b)$ be the set of functions defined on the interval $[a,b]$
 - ★ Note that if $f(x), g(x) \in C(a,b)$ then $\alpha f(x) \in C(a,b)$ and $f(x) + g(x) \in C(a,b)$
- Bounded vectors in $\mathbb{R}^n$ **don't** form a vector space

Outline

1. Vector Spaces
2. **Metrics (distances)**
3. Norms



Metrics

- Vector spaces become more interesting if we have a notion of distance■
- We say $d(x, y)$ is a **distance** or **metric** if
 1. $d(x, y) \geq 0$ (non-negativity)■
 2. $d(x, y) = 0$ iff $x = y$ (identity of indiscernibles)■
 3. $d(x, y) = d(y, x)$ (symmetry)■
 4. $d(x, y) \leq d(x, z) + d(z, y)$ (triangular inequality)■
- There are typically many possible distances (e.g. Euclidean distance, Manhattan distance, etc.)■
- If one or more of the conditions aren't satisfied we have a **pseudo-metric**■

Mappings and Functions

- A function defines a mapping from one vector space to another (although the spaces might be the same), e.g.

$$f : \mathbb{R} \rightarrow \mathbb{R}$$

(f maps the reals to the reals, i.e. $f(x)$ takes a real x and gives you a new real number $y = f(x)$)

- We are often interested in functions that behave nicely
- E.g. They are continuous

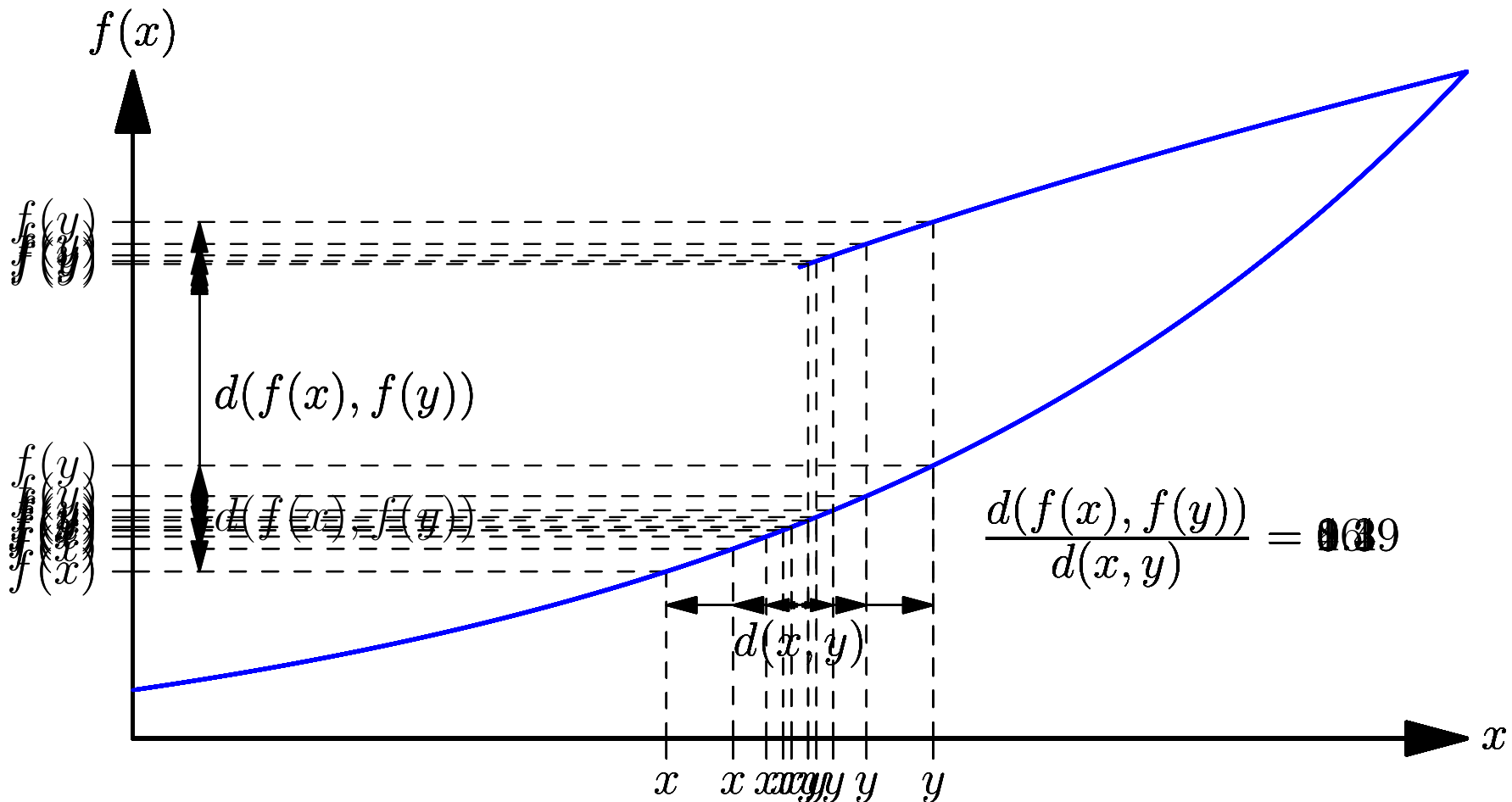
Lipschitz Function

- One way to characterise a well-behaved function, $f(x)$, is if there exists a number $K < \infty$ such that for all x and y

$$d(f(x), f(y)) \leq K d(x, y)$$

- This is known as a **Lipschitz condition** and the function is said to be K -Lipschitz or Lipschitz continuous
- Note that such functions cannot have any jumps (i.e. they are continuous)
- The size of K measures the limit on the amplifying effect of the function

Lipschitz Continuous Functions



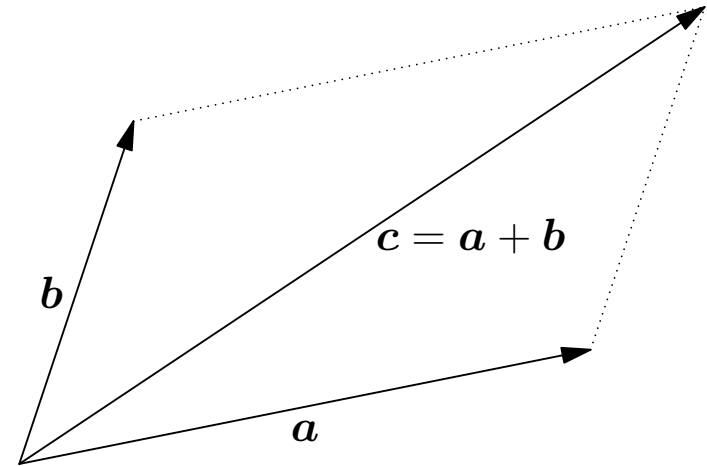
- The definition applies to any function defined in a metric space

Contractive Mappings

- An interesting class of functions are those for which $K < 1$ ■
- These are said to be contractive mappings■
- A famous theorem that applies to contractive mappings is the Banach fixed-point theorem which says there exists a unique fixed point such that $f(x) = x$, and repeatedly applying f converges to it■
- This is used for example in showing that various algorithms will converge■

Outline

1. Vector Spaces
2. Metrics (distances)
3. **Norms**



Norms

- Vector spaces are even more interesting with a notion of length■
- **Norms** provide some measure of the size of a vector■
- To formalise this we define the **norm** of an object $\mathbf{v}$ as $\|\mathbf{v}\|$ satisfying
 1. $\|\mathbf{v}\| > 0$ if $\mathbf{v} \neq \mathbf{0}$ and $\|\mathbf{0}\| = 0$ (non-negativity)■
 2. $\|a\mathbf{v}\| = |a|\|\mathbf{v}\|$ (“linearity”)■
 3. $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ (triangular inequality)■
- When some criteria aren’t satisfied we have a **pseudo-norm**■
- Norms provide a metric $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$ (they are metric spaces)■

Vector Norms

- The familiar vector norm is the (Euclidean) two norm

$$\|\mathbf{v}\|_2 = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

- Other norms exist, such as the p -norm ($p \geq 1$)

$$\|\mathbf{v}\|_p = \left(\sum_{i=1}^n |v_i|^p \right)^{1/p}$$

- Special cases include the 1-norm and the infinite norm

$$\|\mathbf{v}\|_1 = \sum_{i=1}^n |v_i| \quad \|\mathbf{v}\|_\infty = \max_i |v_i|$$

- The 0-norm is a pseudo-norm as it does not satisfy condition 2

$$\|\mathbf{v}\|_0 = \text{number of non-zero components}$$

Norms for Regularisation

- Why so many norms? Here is one place where the choice matters■
- When we fit a model many different weight vectors $\mathbf{w}$ explain the training data equally well■
- We usually prefer a “small” $\mathbf{w}$, so we minimise

$$\text{loss}(\mathbf{w}) + \eta \|\mathbf{w}\| \blacksquare$$

- Equivalently we minimise the loss subject to $\|\mathbf{w}\| \leq c$, confining $\mathbf{w}$ to a **ball** of whichever norm we chose■
- Each norm means something different by “small”■
- $\|\mathbf{w}\|_2$ small: no one component is large, the weight is spread out■
- $\|\mathbf{w}\|_1$ small: a few large components are tolerated provided the rest are tiny■
- $\|\mathbf{w}\|_\infty$ small: only the largest component is controlled■
- **Choosing a norm is choosing what you mean by “small”■**

Matrix Norms

- We can define norms for other objects■
- The norm of a matrix encodes how large the mapping is■
- The Frobenius norm is defined by

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |A_{ij}|^2} \blacksquare$$

- Many other norms exist including 1-norm, max-norm, etc.■
- Some, but not all, norms satisfy the inequality

$$\|\mathbf{AB}\| \leq \|\mathbf{A}\| \times \|\mathbf{B}\| \blacksquare$$

Compatible Norms

- A vector and matrix norm are said to be compatible if

$$\|\mathbf{M}\mathbf{v}\|_b \leq \|\mathbf{M}\|_a \times \|\mathbf{v}\|_b$$

(Spectral and Euclidean norms are compatible)■

- Norms provide quick ways to bound the maximum growth of a vector under a mapping induced by the matrix■
- We will see that a measure of the sensitivity of a mapping is in terms of the ratio of its maximum effect to its minimum effect on a vector■
- This is known as the **conditioning**, given by $\|\mathbf{M}\| \times \|\mathbf{M}^{-1}\|$ ■

Why Should You Care?

- Deep learning involves multiplying the input (which we can think of as a vector $\mathbf{x}$) by many layers■
- In CNNs we have convolutional layers and dense layers■
- The effect of applying these layers can be represented by a matrix multiplication $\mathbf{x}_n = \mathbf{L}_n \mathbf{x}_{n-1}$ ■
- We also do other things like applying ReLUs or pooling that change the magnitude, $\|\mathbf{x}_n\|$, of our representation■
- If you are developing new architectures you want $\|\mathbf{x}_n\|$ neither to blow up nor vanish■
- This can be controlled by carefully choosing $\|\mathbf{L}_n\|$ ■

The Lipschitz Constant of a Network

- We can now put the Lipschitz condition and the matrix norm together
- Recall f is Lipschitz with constant K if

$$\|f(\mathbf{x}) - f(\mathbf{y})\| \leq K \|\mathbf{x} - \mathbf{y}\|$$

- If we compose two functions the constants **multiply**

$$K_{f \circ g} \leq K_f K_g$$

- A layer $\mathbf{x} \mapsto \mathbf{L}_n \mathbf{x}$ has $K = \|\mathbf{L}_n\|_2$, the **spectral norm**—this is just compatibility, $\|\mathbf{L}\mathbf{v}\| \leq \|\mathbf{L}\|_2 \|\mathbf{v}\|$
- ReLUs and max-pooling never stretch distances (though they can shrink them): $K = 1$
- A network is just a composition of layers, so

$$K_{\text{net}} \leq \prod_n \|\mathbf{L}_n\|_2$$

Why Care About K ?

- K tells us how much the output can move when we nudge the input

$$\|f(\mathbf{x} + \boldsymbol{\delta}) - f(\mathbf{x})\| \leq K \|\boldsymbol{\delta}\|$$

- **Adversarial examples** are tiny $\boldsymbol{\delta}$ that change the classification: a network with small K cannot be fooled this easily
- It is the same quantity that controls the signal: $K \gg 1$ and signals (and gradients) explode, $K \ll 1$ and they vanish
- Wasserstein GANs require the critic to have $K \leq 1$. One way to enforce this is **spectral normalisation**: divide each $\mathbf{L}_n$ by $\|\mathbf{L}_n\|_2$
- Bounds on how well a network generalises also depend on K
- But there is a trade-off: make K too small and the network is too smooth to fit the data

Function Norms

- Functions can also have norms, for example, if $f(x)$ is defined on some interval $\mathcal{I}$

$$\|f\|_{L^2} = \sqrt{\int_{x \in \mathcal{I}} f^2(x) dx}$$

- The L^2 vector space is the set of functions for which $\|f\|_{L^2} < \infty$
- The L^1 -norm is given by $\|f\|_{L^1} = \int_{x \in \mathcal{I}} |f(x)| dx$
- The infinite-norm is given by $\|f\|_{\infty} = \max_{x \in \mathcal{I}} |f(x)|$

Norms and Metrics

- Given a norm, there exists an induced distance

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$$

- Any normed space has an associated metric space
- Thus we can define the Frobenius distance between matrices

$$d_F(\mathbf{A}, \mathbf{B}) = \|\mathbf{A} - \mathbf{B}\|_F$$

- It is easy to see that non-negativity, symmetry and the triangular inequality for distances are all inherited from the same properties of norms

Summary

- Vector spaces with a distance (metric spaces) and vector spaces with a norm (normed vector spaces) are interesting objects■
- They allow you to define a topology (open/closed sets, etc.)■
- You can build up ideas about connectedness, continuity, contractive maps, fixed-point theorems, . . . ■
- For the most part we are going to consider an even more powerful vector space that has an inner-product defined■